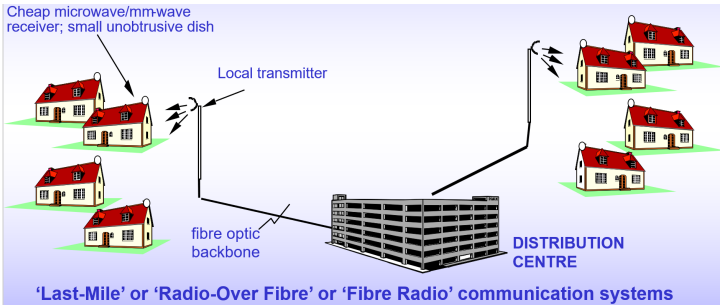
The Solutions for EE4.18 and AO6, 2021

Model answer to Q 1(a): Bookwork and Discussions in Class

Consider a 'Fibre Radio' communications system:

i) Illustrate and briefly explain the basic concept.



The basic idea of Fibre Radio is to effectively replace the TV antenna with a millimetre-wave receiver, having a line-of-sight link to a high-speed transmitter that services a small cluster of houses.

[2]

ii) What is the main reason for introducing this technology?

The overall costs of the millimetre-wave infrastructure may be cheaper than the high costs of digging-up roads to lay down optical cables to each home.

[1]

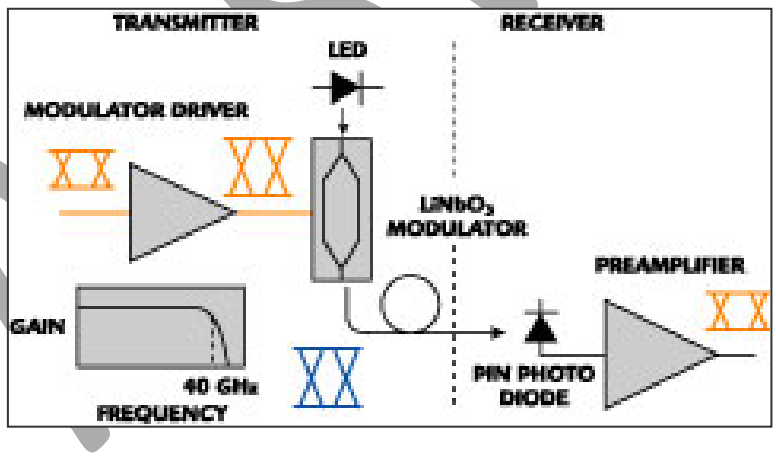
iii) Briefly explain the reasons why its implementation has not been successful.

The overall costs of the millimetre-wave infrastructure is more expensive than the high costs associated with digging up the roads.

[1]

Model answer to Q 1(b): Bookwork

Draw the basic block diagram for a high speed optical communications system, clearly indicating all the blocks.



[3]

Model answer to Q 1(c): Bookwork and Discussions in Class

For the implementation in 1(b):

iv) What are the main systems level challenges?

High Voltage Gain-Bandwidth-Product.

[1]

v) What is the main circuit level challenge?

High Breakdown Voltage.

Answers

[1]

vi) With suitable examples, briefly explain the transistor technology choice as data rates increase from 10 Gb/s to 40 Gb/s.

The lower data rates are generally going to be more ubiquitous and so cost has to be more important than performance for higher data rates. As a result, MESFET technology is relatively cheap and will operate at 10 Gb/s data rates. However, for faster speeds more expensive exotic transistor technologies are needed, such as pseudomorphic-HEMT transistor technologies.

[2]

Model answer to Q 1(d): Bookwork, Discussions in Class and Computed Example

b) For a typical optical driver in 1(b):

i) Input and output voltages are 0.4 V_{pk-pk} and 8 V_{pk-pk}, respectively, and it consumes 2.5 W of DC power. With an output power of 20 dBm, calculate the gain in dB, basic output efficiency and PAE for this power amplifier.

Power gain is 20 log(8/0.4) = 26 dB; Output power is 100 mW and so basic output efficiency is 100x(0.1/2.5) = 4%; Input power is 20 dBm – 26 dB = -6 dBm or 0.25 mW and so PAE is 100x(0.1-0.00025)/2.5 = 3.99%.

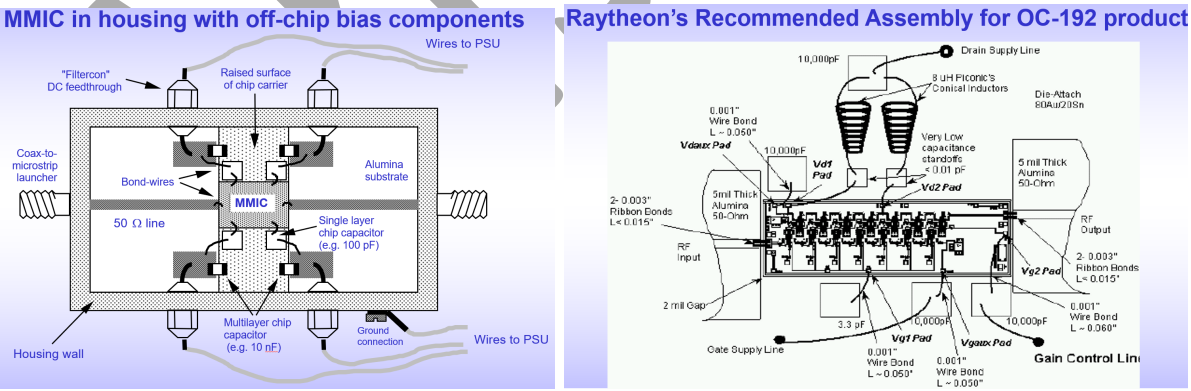
[3]

ii) Briefly explain why an 'all transistor' solution is needed and how this differs from a conventional MMIC amplifier circuit design.

An 'all transistor' solution does not contain any discrete inductors (and ideally no capacitors). As a result, it is less likely to suffer from resonances that could cause ripples in the amplifier's gain-frequency response. With a conventional MMIC, spiral inductors are commonplace, used in DC biasing chokes and impedance matching networks.

[2]

iii) With the use of illustration(s), explain why extreme RF decoupling is needed and how it is realized in the biasing circuitry.



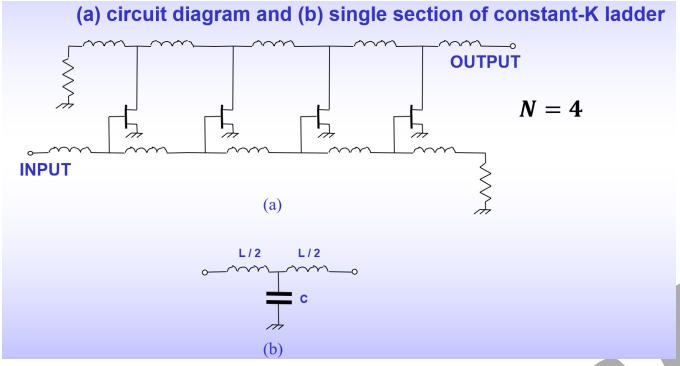
As can be seen in both illustrations, an individual DC bias path to any transistor consists of a low-pass filter, consisting of series inductances (wires, bond wires and chokes) and shunt capacitances (bare-chip, SMD, feedthrough and pads). The spread of values and implementations give more resilience to resonances and ensure a very low cut-off frequency. This helps to facilitate ultra-broadband operation, especially close to DC.

[4]

Answers

Model answer to Q 2(a): Bookwork and Discussion in Class

Draw the basic circuit diagram for a 4-stage distributed amplifier (also known as the travelling-wave amplifier). Describe the basic principle of operation and from this state its main advantage over traditional amplifiers.



The artificial gate transmission line absorbs Cgs, to create a Lg-Cgs ladder network; while the drain line absorbs Cds to create an Ld-Cds ladder network. As a result, there should ideally be no resonances, being ideal for ultra-wideband applications. The input gate line must have a phase velocity that matches the output drain line.

[5]

Model answer to Q 2(b): Computed Example

Given identical FETs, represented by a simple equivalent circuit model having intrinsic values of C_{gs} = 1 pF, C_{ds} = 0.5 pF and g_m = 100 mS, calculate the inductance values and cut-off frequency for a 50 Ω input gate transmission line. Also, for a 4-stage distributed amplifier, calculate the available power gain. For a constant-K ladder network, the phase velocity per unit section is given by Equation 2.1 and the available power gain is given by Equation 2.2:

$$Velocity/Section = \frac{1}{\sqrt{LC}} \tag{2.1}$$

$$Available\ Power\ Gain = \left(\frac{N g_m \sqrt{Z_{og} Z_{od}}}{2}\right)^2 \tag{2.2}$$

$C_{gs} = 1\,pF \qquad C_{ds} = 0.5\,pF$

$Z_{og} = \sqrt{\frac{L_g}{C_{gs}}} = 50\,\Omega$

$\therefore L_g = 2.5\,nH$

$\frac{V_{pg}}{Section} = \frac{1}{\sqrt{L_g C_{gs}}} = \frac{V_{pd}}{Section} = \frac{1}{\sqrt{L_d C_{ds}}}$

$\therefore \frac{C_{gs} L_g}{C_{ds} L_d} = 1 \quad \therefore L_d = 2L_g = 5\,nH$

$Z_{od} = \sqrt{\frac{L_d}{C_{ds}}} = 100\,\Omega$

$f_c = \frac{1}{\pi\sqrt{LC}} = 6.366\,GHz$ FOR BOTH GATE AND DRAIN LINES

AVAILABLE POWER GAIN $= \left(\frac{4 g_m \sqrt{Z_{og} Z_{od}}}{2}\right)$

$= 200 \equiv 23\,dB$

[5]

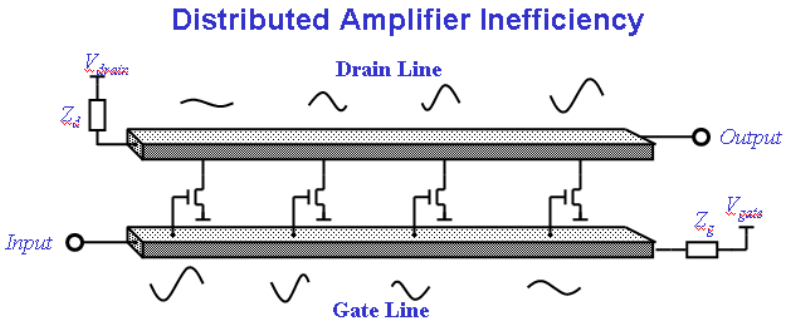
Answers

Model answer to Q 2(c): Bookwork and Discussion in Class

Explain why the gain-bandwidth product does no theoretically change if the size of the transistors in 2(b) is doubled. Also, explain why more stages do not necessarily give more power gain.

Doubling transistor size:

- Doubles both Cgs and Cds and so:
- Reduces Zo by a factor of √2 but also doubles gmo and so increases available power gain by 3 dB
- Decreases the Bragg frequencies by a factor of √2
- Therefore, the voltage gain-bandwidth product is constant!



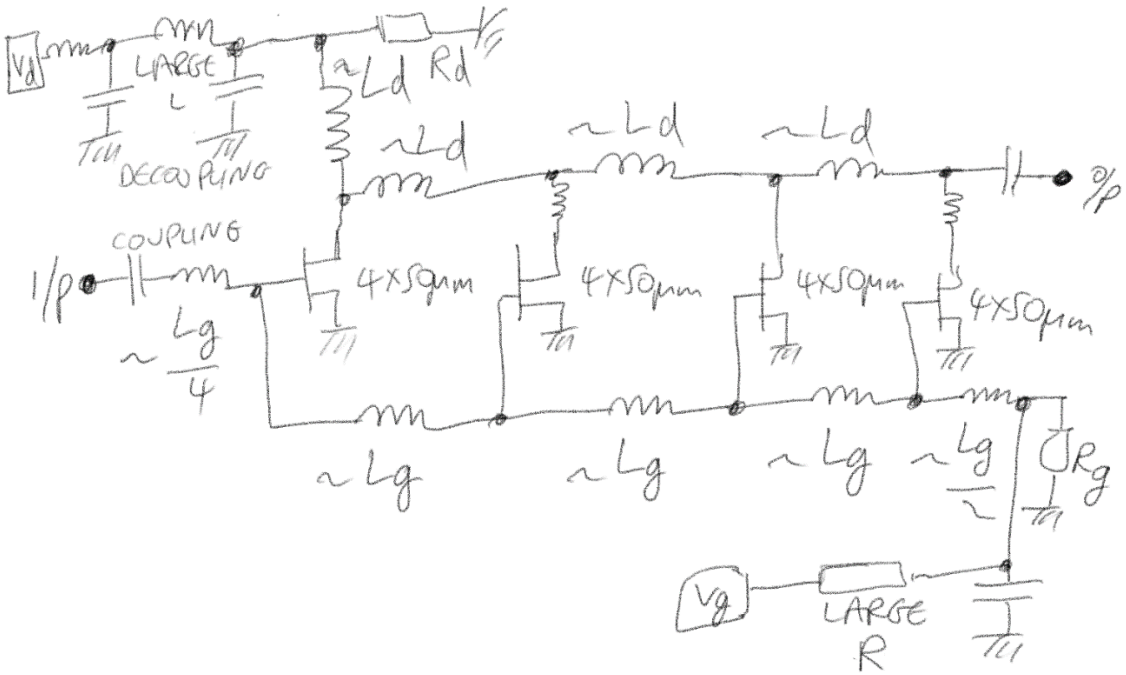
- Power flow in both directions
- Unequal voltages along the drain line
- Imperfect termination
- Losses in gate and drain line

When you have more stages the losses in the gate line attenuate the signal so much as it passes down the line that the associated amplifiers have difficulty amplifying it. As a result, there is little to be gained. Therefore, after about 4 stages, there if diminishing returns on having more stages.

[5]

Model answer to Q 2(d): New Application

From the photograph in Figure 2.1, using deduction, draw the equivalent circuit model.



[5]

Model answer to Q 3(a): Computed Example

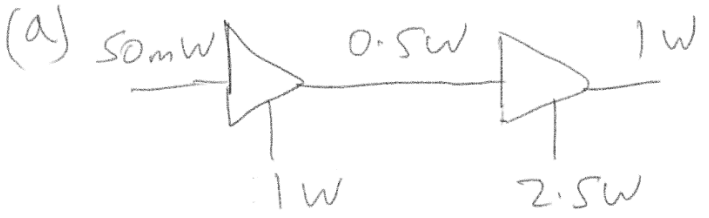
With the normal driver-PA configuration, calculate:

i) Overall gain. [1]

ii) Overall Basic Output Efficiency and PAE. [3]

iii) Overall Noise Figure. [2]

iv) Overall IP3. [2]



(i) $G = 10 + 3 = 13\,dB$

(ii) $\eta_1 = \frac{0.5}{1} = 50\%$

$\eta_2 = \frac{1}{2.5} = 40\%$

$\eta = \eta_1 G_1 // \eta_2$

$= \frac{5 \times 0.4}{5 + 0.4} = 37.0\%$

$\eta_{PAE} = \eta\left(1 - \frac{1}{G}\right) = 35.15\%$

(iii) $F = F_1 + \frac{F_2 - 1}{G_1}$

$= 1.585 + \frac{3-1}{10} = 1.785$

$\equiv 2.52\,dB$

(iv) $IP_3 = (100{,}000 \times 2) // 10{,}000$

$= \frac{200{,}000 \times 10{,}000}{210{,}000}$

$\equiv 49.79\,dBm$

Model answer to Q 3(b): Textbook Derivation

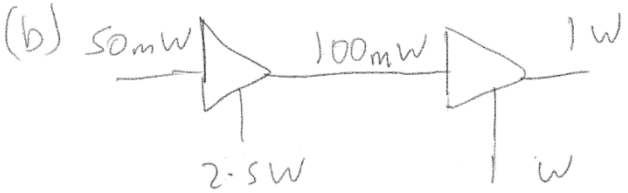
With the PA mistakenly feeding the driver, calculate:

v) Overall gain. [1]

vi) Overall Basic Output Efficiency and PAE. [3]

vii) Overall Noise Figure. [2]

viii) Overall IP3. [2]



(i) $G = 10 + 3 = 13\,dB$

(ii) $\eta_1 = \frac{0.1}{2.5} = 4\%$

$\eta_2 = \frac{1}{1} = 100\%$

$\eta = \eta_1 G_1 // \eta_2$

$= \frac{0.8 \times 1}{0.8+1} = 44.4\%$

$\eta_{ADD} = 2\left(1 - \frac{1}{G}\right) = 42.22\%$

(iii) $F = F_1 + \frac{F_2 - 1}{G_1}$

$= 2 + \frac{1.585 - 1}{2} = 2.295$

$= 3.60\,dB$

(iv) $IP_3 = (10{,}000 \times 10) // 100{,}000$

$= \frac{100{,}000^2}{200{,}000} = 50{,}000$

$= 46.99\,dBm$

Model answer to Q 3(c): Deductive Reasoning

Comment on the output power levels for both stages, relative to P_{1dB}, with both 3(a) and 3(b). What would you expect the power gain to do with each stage in 3(b)?

With 3(a), the output power of the first stage is at the 1 dB compression point, while that of the second stage is beyond this and, therefore, closer to the saturation pint where gain is compressed.

With 3(b), the output power level for the first stage is within the linear operation region and so the gain is expected to be greater than 3 dB. Conversely, the output power level of the second stage is greater than the 1 dB compression point and so its gain is expected to be less than 10 dB.

[3]

Model answer to Q 3(d): New Application of Theory

With 3(b), is the basic output efficiency of the second stage practical and if not state why?

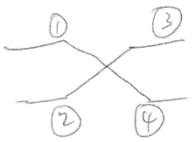
The second stage has a basic output efficiency of 100%, which is not theoretically possible. Moreover, when driven closer to saturation, the output impedance requirements will be below 50 ohms and so there will be additional mismatch loss.

[1]

Model answer to Q 4(a): New application of theory

An ideal four-port coupled-line directional coupler has the following S-parameters: $S_{21} = 0.6$ to its coupled port and $S_{41} = -j0.8$ to its direct port. Draw a symbolic representation of this coupler, labelling the appropriate port numbers. By inspection, write down the complete S-parameter matrix for this coupler, identifying all assumptions about port impedance matching, symmetry and reciprocity.

All ports are assumed to be lossless (with perfectly impedance matching at all ports), while the coupler is symmetrical and reciprocal.



$S_{21} = 0.6$

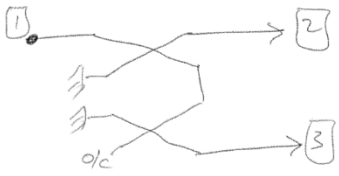
$S_{41} = -j0.8$

$$[S] = \begin{pmatrix} 0 & 0.6 & -j0.8 & 0 \\ 0.6 & 0 & 0 & -j0.8 \\ -j0.8 & 0 & 0 & 0.6 \\ 0 & -j0.8 & 0.6 & 0 \end{pmatrix}$$

[4]

Model answer to Q 4(b): Bookwork and Discussion in Class

Using the symbolic representation for the ideal coupler in 4(a), draw the circuit schematic for an ideal three-port Marchand balun, labelling the port numbers.



[3]

Model answer to Q 4(c): New Derivation and Computed Example

By inspection of 4(b), derive the expression of S_{21} for the balun and calculate this coefficient.

All Versions

$$S_{21} = S_{21}(-1)S_{41} + S_{41}S_{41}\,1\,S_{41}S_{21} + S_{41}S_{41}\,S_{21}(-1)S_{21}\,S_{41}S_{21} + S_{41}S_{41}\,S_{21}(-1)S_{21}\,S_{21}(-1)S_{21}\,S_{41}S_{21} \ldots$$

$$\sum_{n=0}^{\infty} \rho^n \equiv \frac{1}{1-\rho} \quad \text{WHERE } \rho = -S_{21}^2$$

$$= 0.73529$$

$$\therefore S_{21} = -S_{21}S_{41} + S_{21}S_{41}^3(0.73529)$$

$$= +j0.48 + j0.22588$$

$$= +j0.70588 = j/\sqrt{2}$$

[6]

Model answer to Q 4(d): New Derivation and Computed Example

By inspection of 4(b), derive the expression of S_{31} for the balun and calculate this coefficient. The same use of Equation (4.1) will be needed. Using the results here and from 4(c), confirm that this circuit operates as a balun.

$$S_{31} = S_{41}S_{21} + S_{41}S_{41}S_{21}(-1)\,S_{41} + S_{41}S_{41}S_{21}(-1)\,S_{21}S_{21}(-1)\,S_{41} + S_{41}S_{41}S_{21}(-1)\,S_{21}S_{21}(-1)\,S_{21}S_{21}(-1)\,S_{41} \ldots$$

$$\therefore S_{31} = S_{21}S_{41} - S_{21}S_{41}^3(0.73529)$$

$$= -j/\sqrt{2} \equiv -S_{21}$$

$\therefore$ -3dB AT BOTH OUTPUTS, BEING IN ANTIPHASE

[4]

Model answer to Q 4(e): New Derivation and Computed Example

By inspection of 4(b), derive the expression of S_{11} for the balun and calculate the input return loss. The same use of Equation (4.1) will be needed.

$$S_{11} = S_{21}(-1)S_{21} + S_{41}S_{41}(1)\,S_{41}S_{41} + S_{41}S_{41}(1)\,S_{21}(-1)S_{21}\,S_{41}S_{41} + S_{41}S_{41}(1)\,S_{21}(-1)S_{21}\,S_{21}(-1)S_{21}\,S_{41}S_{41}$$

$$\therefore S_{11} = -S_{21}^2 + S_{41}^4(0.73529)$$

$$= -0.36 + 0.3012$$

$$= -0.0588$$

$\therefore$ RETURN LOSS $= 24.6$ dB

[3]

Model answer to Q 5(a): New Application and Computed Example

Calculate the Maximum Unilateral Transducer Power Gain, given by Equation (5.1):

$$G_{TU_max} = \frac{1}{1-|S_{11}|^2}|S_{21}|^2\frac{1}{1-|S_{22}|^2} \quad (5.1)$$

$G_{TU-MAX} = \frac{1}{1-|S_{11}|^2}|S_{21}|^2\frac{1}{1-|S_{22}|^2}$

$|S_{11}| = 0.5 \quad |S_{21}| = 20 \quad |S_{22}| = 0.4$

$G_{TU-MAX} = 1.333 \times 20 \times 1.1905$

$= 31.745 \equiv 15\,dB$

[4]

Model answer to Q 5(b): Bookwork and Computed Example

Using textbook equations, calculate the Rollet Stability Factor. Is the amplifier unconditionally or conditionally stable and why?

ROLLET STABILITY FACTOR

$$K = \frac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|}$$

$\Delta = S_{11}S_{22} - S_{12}S_{21}$

$\therefore K = 2.22$ UNCONDITIONALLY STABLE

since K > 1

[6]

Model answer to Q 5(c): Bookwork and Computed Example

Depending on your answer in 5(b), using textbook equations, calculate either the Maximum Stable Gain or Maximum Available Gain in dB. How does this compare with the value in 5(a)?

MAXIMUM AVAILABLE GAIN

$MAG = \left|\frac{S_{21}}{S_{12}}\right|\left(K-\sqrt{K^2-1}\right)$

SINCE $K > 1$

$\therefore MAG = \frac{20}{0.2}(0.260)$

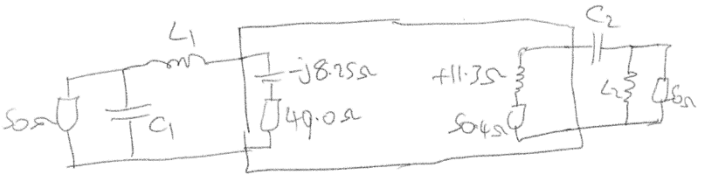
$\therefore MAG = 26 \equiv 14.15\,dB$

THIS IS JUST BELOW G_{TU-MAX}

[4]

Model answer to Q 5(d): New Application and Computed Example

The input impedances looking into the transistor is 49.0 - j8.25 Ω. Using discrete elements, design a two-element impedance matching network. What is the ideal -3 dB bandwidth of this input matching stage?



$$Q_1 = \sqrt{\left(\frac{50}{49}\right) - 1} = 0.1429$$

$$\omega C_1 z_0 = Q_1 \therefore C_1 = 185.7\,fF$$

$$\frac{\omega L_1'}{49} = Q_1 \therefore L_1' = 454.9\,pH$$

$$\omega L_1'' = 8.25\,\Omega \therefore L_1'' = 535.9\,pH$$

$$\therefore L_1 = L_1' + L_1'' = 991\,pH$$

$$\text{LOADED-}Q = \frac{f_0}{BW_{3dB}}$$

$$\therefore BW_{3dB} = 0.1429 \times 2.45\,GHz = 350\,MHz$$

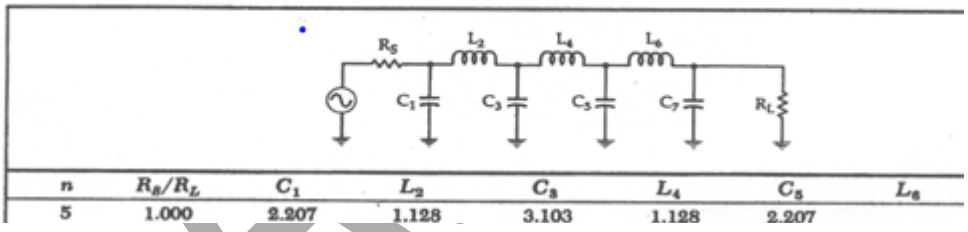
[6]

Model answer to Q 6(a): Computed Example

The 802.11 b/g/n Wi-Fi allocated spectrum is from 2401 MHz to 2484 MHz.

Block out the 802.11 b/g/n Wi-Fi signal by designing a suitable 5th-order Chebyshev filter, having a worst-case passband ripple of 1 dB, with shunt-first element and 50 Ω port terminations.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple



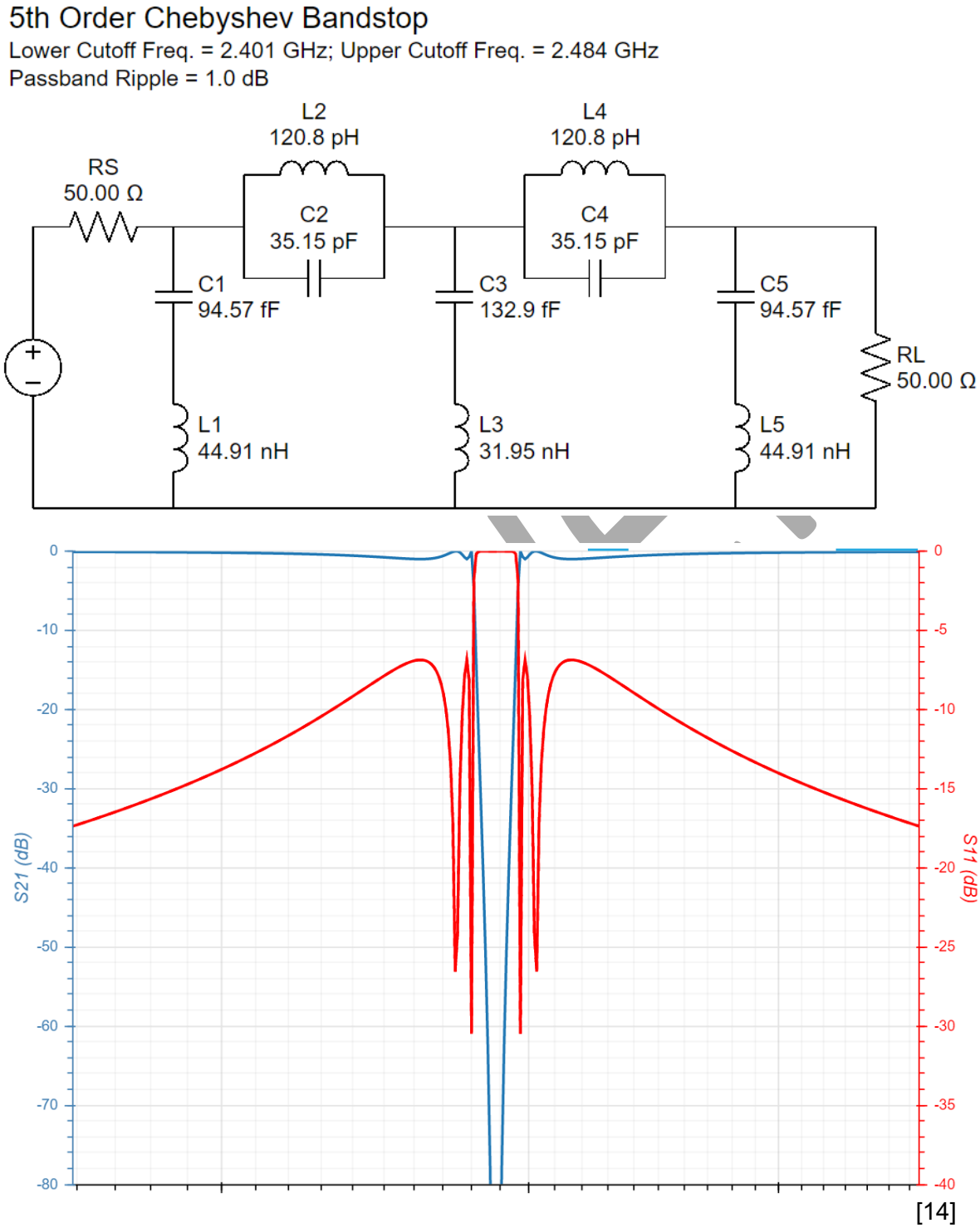
n	R_s/R_L	C_1	L_2	C_3	L_4	C_5	L_6
5	1.000	2.207	1.128	3.103	1.128	2.207	

For a band stop filter, the capacitors are replaced by a series *LC* resonators and the inductors are replaced by parallel *LC* resonators. The un-normalized shunt connected series tuned circuit element values are:

$$Cs = \frac{BpCn}{2\pi f_o^2 R_L} \quad and \quad Ls = \frac{R_L}{2\pi BpL_n}$$

The un-normalized series connected parallel tuned circuit element values are:

$$Cp = \frac{1}{2\pi BpCnR_L} \quad and \quad Lp = \frac{BpLnR_L}{2\pi f_o^2}$$



[14]

Model answer to Q 6(b): Computed Example

With the design in 6(a), what is the level of attenuation at the band edges and at 2421.75 MHz?

At the band edges, the attenuation is 3 dB.

Passband bandwidth, Bp = f_{P1}- f_{P2} = 83 MHz.

Stopband bandwidth, Bs = 2 x (2421.75 MHz - 2401 MHz) = 41.5 MHz
Therefore, f/fc = Bp/Bs = 2. From the attenuation curves, the attenuation is 48 dB.

[3]

Model answer to Q 6(c): Computed Example

Calculate the worst-case level of passband return loss?

From the conservation of energy principle: $1 = |S_{11}|^2 + |S_{21}|^2$

With a worst-case passband ripple of 1 dB, $|S_{21}|^2 = 0.7943$

Therefore, $|S_{11}|^2 = 0.2057$, giving a return loss of 6.868 dB

[3]

Answers